Journal of Green Economy and Low-Carbon Development
https://www.acadlore.com/journals/JGELCD

Digital Economy Development and Manufacturing Carbon Emissions: Industrial Restructuring and Regional Heterogeneity in the Guangdong–Hong Kong–Macao Greater Bay Area

Yuanhua Yang[1*], Yao Xiao[2], Dengli Tang[2]

[1] School of Finance, Guangdong University of Finance and Economics, 510320 Guangzhou, China
[2] School of Business Administration, Guangdong University of Finance and Economics, 510320 Guangzhou, China

* Correspondence: Yuanhua Yang (yuanhuayang@gdufe.edu.cn)

Received: 01-13-2026 **Revised:** 02-23-2026 **Accepted:** 03-05-2026

Citation: Y. H. Yang, Y. Xiao, and D. L. Tang, "Digital economy development and manufacturing carbon emissions: Industrial restructuring and regional heterogeneity in the Guangdong–Hong Kong–Macao Greater Bay Area," *J. Green Econ. Low-Carbon Dev.*, vol. 5, no. 1, pp. 18–41, 2026. https://doi.org/10.56578/jgelcd050102.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Digital economy development changes manufacturing production and its energy requirements, but the carbon implications may differ across cities. This study examined nine mainland cities in the Guangdong–Hong Kong–Macao Greater Bay Area over 2011–2024 using fixed-effects models, a spatial Durbin model (SDM), mediation regressions, and descriptive decoupling analysis. The reported SDM coefficient for digital economy development was negative (−0.3423, standard error 0.1631), whereas the decomposed average direct and total effects were statistically insignificant. In the non-spatial mediation model, the indirect association through the secondary-industry share represented 29.82% of the estimated total association, while the research and development (R&D) expenditure pathway was not statistically significant. The spatial indirect effect was also insignificant. Subgroup coefficients were negative in Guangzhou, Shenzhen, Foshan, and Dongguan (−0.9511) and positive but insignificant in the five peripheral cities (0.3047). These patterns suggest that digital development and manufacturing emissions have context-dependent relationships. They support evaluating digital investment alongside industrial composition and regional carbon accounting. The findings concern conditional associations in a small regional panel and a direct fuel-combustion inventory; they should be interpreted within those statistical and accounting boundaries.

Keywords: Digital economy; Manufacturing carbon emissions; Industrial restructuring; Spatial Durbin model; Regional heterogeneity; Low-carbon development

1 Introduction

Manufacturing regions face a difficult adjustment: they must sustain production and employment while reducing the carbon intensity and, ultimately, the absolute emissions of industrial activity. Digital technologies change the information available to firms, the organization of production, and the coordination of supply chains. Their environmental consequences depend on the activities they replace, the electricity they require, and the production expansion they make possible. A useful assessment therefore needs to distinguish emissions levels, emissions intensity, and broader measures of carbon efficiency.

Evidence from Chinese cities links digital development to carbon emission performance [1], while Ma et al. [2] examined whether research and development (R&D) investment moderates this relationship. These are related questions, but they concern different outcomes and mechanisms. Evidence on structural deviation [3] and industrial carbon efficiency [4] further suggests that industrial composition matters alongside production technology. Dynamic city-level analysis [5] and research on three major urban agglomerations [6] motivate closer attention to development stages and regional context. These studies provide a basis for examining manufacturing emissions within the Pearl River Delta (PRD) rather than assuming that national estimates apply uniformly to its constituent cities.

The present study concerns the nine mainland cities of the Guangdong–Hong Kong–Macao Greater Bay Area. Hong Kong and Macao form part of the regional setting but are outside the estimation sample. The nine cities share production networks and provincial institutions, while differing in manufacturing scale, specialization, and digital development. This setting allows a focused comparison within an integrated manufacturing region, although the small number of cities limits statistical precision and the scope for generalization.

Research on digital investment and carbon efficiency [7], emissions along the digital economy's industry chain [8], and the interaction between energy structure and digitalization [9] illustrates why a single environmental interpretation is insufficient. Digital investment can influence production efficiency while also generating upstream energy demand. Information and communication technology (ICT) research outside China likewise considers transmission channels and country conditions [10]; earlier evidence on electricity consumption documents the energy demands associated with ICT expansion [11]. The Chinese digital economy also encompasses institutional and structural changes beyond the adoption of individual technologies [12]. Consequently, an aggregate city index should be interpreted as a measure of the wider digital environment rather than a direct measure of factory-level digital adoption.

Manufacturing-specific evidence on digital transformation and carbon competitiveness [13] helps narrow the sectoral question. Research on technological diversification and industrial carbon efficiency [14], together with provincial evidence on digital development and emissions intensity [15], shows the relevance of separating technology, composition, and scale. The present analysis follows that distinction. Its dependent variable is estimated direct fuel-combustion CO_2 from manufacturing; its mechanism variables describe broad industrial composition and aggregate R&D expenditure. These measures support a regional association study, but they do not trace individual technologies or production transfers.

Three questions guide the analysis. First, is digital economy development negatively associated with manufacturing fuel-combustion emissions after accounting for city effects, year effects, and observed covariates? Second, are industrial composition and R&D expenditure statistically consistent with the proposed channels? Third, do spatial impacts and subgroup estimates reveal a common regional pattern or contrasting relationships across core and peripheral cities?

A panel covering 2011–2024 was examined using fixed-effects and spatial models, separate mediation regressions, and descriptive decoupling analysis. The study examines whether the association between digital development and manufacturing emissions differs across cities within an integrated regional economy. It considers industrial composition and R&D expenditure as separate potential channels and compares local and cross-city associations using a spatial framework. Section 2 develops the analytical framework. Section 3 presents the data and methods. Section 4 reports results, Section 5 discusses their interpretation and policy implications, and Section 6 concludes.

2 Literature Review and Hypothesis Development

2.1 Scale, Composition, and Technology

Environmental economics distinguishes changes in production scale, economic composition, and production techniques. The trade–environment framework of Copeland and Taylor [16] and the growth analysis of Grossman and Krueger [17] provide foundations for this distinction. Applied to digital development, the scale channel concerns additional production; the composition channel concerns shifts among activities; and the technique channel concerns the emissions required to produce a given output. These channels can operate simultaneously and in opposing directions.

Directed technical change adds a further qualification: innovation responds to incentives and need not automatically favour cleaner technologies [18]. Digital applications may improve monitoring and equipment operation, but they can also support more production in emissions-intensive activities. Accordingly, the empirical hypothesis concerns the net conditional association in the sample, rather than a universal technological rule.

H1. Digital economy development is negatively associated with manufacturing carbon emissions, conditional on the specified city characteristics and fixed effects.

2.2 Industrial Restructuring

Digital connectivity can reduce coordination costs between suppliers and customers, making it easier to organize production across locations. It may also accompany the expansion of information services, design, and other activities that change the composition of city output. The literature on structural deviation [3] and manufacturing carbon efficiency [4] provides relevant empirical context, but neither broad structural adjustment nor a larger service sector necessarily means that existing manufacturing plants have become cleaner.

This study measures industrial structure (IND) using secondary-industry value added as a percentage of GDP. A decline in this share could arise through service-sector growth, a contraction of secondary industry, or changes in relative prices. Secondary industry also includes activities beyond manufacturing. The variable therefore represents broad economic composition. It does not directly measure the replacement of high-carbon manufacturing subsectors by low-carbon subsectors or the relocation of particular plants.

H2. Changes in the secondary-industry share form a statistical channel in the negative association between digital economy development and manufacturing carbon emissions.

2.3 R&D Expenditure and Cleaner Production

Digital systems may support experimentation, simulation, and the acquisition of technical information. These capabilities make an R&D channel plausible, but expenditure is an input measure rather than evidence of successful environmental innovation. Research spending may support new products, expansion, or process improvements unrelated to energy use. Conversely, firms can adopt cleaner equipment developed elsewhere without a corresponding increase in local R&D expenditure.

The distinction matters when comparing manufacturing regions. Digital transformation and carbon competitiveness [13] concern a different outcome from expenditure-based mediation. Technological diversification and specialization [14] likewise describe innovation characteristics that an aggregate spending measure cannot capture. The hypothesis below is therefore evaluated using the stated proxy, and its interpretation remains specific to that proxy.

H3. R&D expenditure forms a statistical channel in the negative association between digital economy development and manufacturing carbon emissions.

2.4 Decoupling and the Energy Requirements of Digitalization

Tapio's framework [19] classifies the relationship between changes in emissions and economic activity. Strong decoupling means falling emissions alongside increasing output; weak decoupling allows emissions to rise more slowly than output. These states describe observed trajectories. Their occurrence does not identify which policy or technology produced them.

The digital sector itself has a carbon footprint. Reviews of ICT emissions [20] and evidence on data-centre energy use [21] show the importance of system boundaries, energy efficiency, and changing demand. An apparent improvement in a manufacturing fuel-combustion inventory may coexist with rising electricity use. The present decoupling analysis therefore complements the regression evidence but does not measure the complete environmental footprint of digital development.

Decoupling analysis is used as a descriptive complement to the regression results. It characterizes manufacturing output and emissions trajectories without estimating the contribution of digital economy development to individual decoupling transitions.

2.5 Recent Evidence and Regional Contingencies

Recent studies continue to broaden the empirical question. Research on fiscal decentralization and carbon intensity [22] examines institutional conditions; work on heterogeneous carbon-efficiency relationships [23] considers differences across settings; and city-level efficiency analysis [24] examines potential industrial and innovation pathways. Studies of urban emissions [25] and energy transition [26] further place digitalization within a changing energy system. These outcomes should not be treated as interchangeable with the absolute fuel-combustion emissions examined here.

Structural decomposition research [27] is especially relevant because changes in digital production and digital applications can have different upstream and downstream consequences. Evidence on industrial-structure mediation and spatial spillovers [28], spatial relationships across cities [29], and provincial spillover channels [30] motivates a model that allows cross-city dependence without imposing a predetermined spillover sign. Research on internet development and energy consumption [31] provides another reason to distinguish digital connectivity from energy savings.

The literature also connects digital development to green innovation [32], green total factor productivity [33], energy efficiency [34], and firm environmental performance [35]. These strands motivate the choice of mechanisms, but their measures and levels of aggregation differ from the present city panel. Provincial evidence on digital development, technological progress, and emissions [36] offers a closer comparator, although intensity can fall while absolute emissions increase. The contribution of the present study is consequently a focused regional comparison, not a claim that the digital economy–carbon relationship has not previously been studied.

2.6 Spatial Relationships and the Study Framework

Spatial econometrics distinguishes regression parameters from direct, indirect, and total impacts [37]. A city can respond to changes elsewhere through production linkages, shared infrastructure, and feedback. A significant spatial autoregressive parameter does not establish a significant spillover from digital development; that proposition requires the corresponding impact estimate and its uncertainty.

Firm-level research links digital technology and green innovation to manufacturing carbon performance [38]. At the regional scale, studies of interregional CO_2 outsourcing [39] and consumption-based accounting [40] show why the location of production must be distinguished from the location of final demand. Research on digitalization, IND, and green economic growth [41] also motivates the composition channel considered here. Together, these

studies suggest that lower emissions recorded in one city need not imply an equivalent reduction across the regional production network.

H4. Digital economy development is associated with a non-zero cross-city spillover in manufacturing carbon emissions. The direction is left open because production relocation and cleaner-technology diffusion can imply different signs.

The core–periphery comparison is an additional exploratory analysis. A negative coefficient in one group and an insignificant coefficient in another do not by themselves establish a significant difference between groups. Likewise, observational mediation requires assumptions beyond those needed to estimate ordinary conditional associations [42, 43]. Research on digitalization and energy consumption [44] and interregional digital-economy linkages [45] reinforce the need to keep the proposed mechanisms, observed proxies, and accounting boundaries aligned. These distinctions guide the interpretation of the results below.

3 Data and Methodology

3.1 Study Area and Data Sources

This study examined nine cities in the PRD, namely Guangzhou, Shenzhen, Zhuhai, Foshan, Huizhou, Dongguan, Zhongshan, Jiangmen, and Zhaoqing. These cities constitute the mainland manufacturing component of the Guangdong–Hong Kong–Macao Greater Bay Area. Hong Kong and Macao were excluded because their statistical systems, economic structures, and manufacturing profiles differ substantially from those of the nine mainland cities. Restricting the analysis to the PRD cities provided a more comparable regional setting for examining the relationship between digital economy development and manufacturing carbon emissions.

The dataset covered the period from 2011 to 2024 and comprised 126 city–year observations. Data were compiled from the municipal statistical yearbooks and annual statistical communiqués of the nine cities (2012–2025 editions), the Guangdong Statistical Yearbook, provincial and municipal energy-intensity statistics, the Peking University Digital Financial Inclusion Index, and meteorological statistical records. Manufacturing energy consumption per unit of output and manufacturing output value were used to estimate carbon dioxide emissions, while economic, industrial, digital infrastructure, and innovation indicators were compiled from the corresponding statistical sources. Appendix A records the source and construction of each variable.

Missing observations were treated using linear interpolation within individual cities where adjacent observations were available. Where interpolation was not feasible, missing values were estimated using growth rates from comparable cities within Guangdong Province. The digital economy index for 2020 was taken from the linearly interpolated version of the source table because the original 2020 values contained a documented anomaly, and the 2024 values were recalculated as described in Section 3.2.2. Variables enter the regressions in constant 2011 prices and in logarithmic form where indicated. No winsorization was applied; the descriptive statistics in Section 3.7 therefore reflect the compiled series.

The resulting panel was used for the descriptive analysis, baseline regressions, spatial econometric estimation, and mechanism tests.

3.2 Variable Definition and Measurement

3.2.1 Manufacturing carbon emissions

Manufacturing carbon emissions constituted the dependent variable. Carbon dioxide emissions were estimated using an energy-consumption-based accounting approach consistent with the general principles of the IPCC Guidelines for National Greenhouse Gas Inventories [46].

Eight energy sources were considered: raw coal, coke, crude oil, gasoline, kerosene, diesel, fuel oil, and natural gas. Energy consumption was first converted into standard coal equivalents using the conversion factors reported in the China Energy Statistical Yearbook. The resulting quantities were combined with the fuel-specific carbon emission coefficients listed in Table 1.

The accounting procedure can be expressed as:

$$\mathrm{CO}_{2,it} = \frac{44}{12} \sum_{k=1}^{8} E_{kit} \delta_k \phi_k \tag{1}$$

where, $\mathrm{CO}_{2,it}$ denotes manufacturing carbon dioxide emissions in city i during year t; E_{kit} is the physical consumption of energy source k; δ_k represents its conversion factor into standard coal equivalents; and ϕ_k is the corresponding carbon emission coefficient expressed in tonnes of carbon per tonne of standard coal. The factor $44/12$ converts carbon mass into carbon dioxide mass.

The municipal statistical yearbooks do not publish fuel-by-fuel physical consumption for the manufacturing sector alone; they report comprehensive energy consumption per unit of manufacturing output in standard-coal terms. The fuel-level sum in Eq. (1) was therefore evaluated at the standard-coal aggregate: $\sum_k E_{kit}\delta_k$ equals total

manufacturing energy consumption, obtained by combining manufacturing output value with comprehensive energy consumption per unit of output, and the aggregate carbon coefficient $\bar{\phi} = 0.7476$ t C per tce—the raw-coal coefficient, consistent with the coal-dominated fuel mix of manufacturing in Guangdong—was applied to that aggregate before the $44/12$ conversion. Weighting the eight fuel-specific coefficients of Table 1 by the actual fuel mix would yield a slightly lower aggregate value, so the emission levels reported here are, if anything, on the high side; because a single factor scales every observation, the estimated elasticities and the relative comparisons across cities and years are unaffected.

Table 1 summarizes the energy conversion factors and carbon emission coefficients used in the emissions inventory.

Table 1. Energy conversion factors and carbon emission coefficients

Energy Source	Standard Coal Conversion Factor	Carbon Emission Coefficient
Raw coal	0.7143	0.7476
Coke	0.9714	0.8550
Crude oil	1.4286	0.5854
Gasoline	1.4714	0.5532
Kerosene	1.4714	0.5714
Diesel	1.4571	0.5913
Fuel oil	1.4286	0.6176
Natural gas	13.300	0.4479

Note: For solid and liquid fuels, conversion factors are expressed in kilograms of standard coal per kilogram of fuel; for natural gas, in tonnes of standard coal per 10,000 m^3, following the China Energy Statistical Yearbook. Carbon emission coefficients are expressed in tonnes of carbon per tonne of standard coal equivalent and follow the values compiled by the Energy Research Institute of the National Development and Reform Commission; the raw-coal coefficient of 0.7476 t C/tce is applied to the standard-coal aggregate as explained above. The factor 44/12 converts carbon mass into carbon dioxide mass. Output values are expressed in constant 2011 prices; the change in the output series from 2020 reflects a documented change in the published statistical caliber and is discussed in the text.

The dependent variable was defined as the natural logarithm of manufacturing carbon emissions:

$$\ln \mathrm{CO}_{2,it} = \ln(\mathrm{CO}_{2,it}) \tag{2}$$

Carbon emission intensity was also calculated to examine emissions relative to manufacturing output:

$$\mathrm{CI}_{it} = \frac{\mathrm{CO}_{2,it}}{\mathrm{GM}_{it}} \tag{3}$$

where, GM_{it} represents manufacturing output value in 10^8 CNY. Manufacturing output value refers to gross manufacturing output for 2011–2019 and to the gross industrial output of above-designated-size enterprises for 2020–2024, reflecting a change in the published statistical caliber; the 2024 value was estimated from the 2023 value and the reported 2024 growth rate of above-designated-size industrial value added. All output values are expressed in constant 2011 prices, and the same series was used for the carbon-intensity measure and the decoupling analysis. Carbon emission intensity was expressed in tonnes of carbon dioxide per CNY 10,000 of manufacturing output.

The emissions inventory followed a production-based accounting boundary. Because comprehensive energy consumption includes purchased electricity and other secondary energy, the series covers energy-related emissions of manufacturing activity within the city. Industrial process emissions that are not embodied in energy consumption, such as those from cement clinker or steel production, were not separately estimated. The estimates therefore represent energy-related manufacturing emissions rather than the complete carbon footprint of manufacturing activity.

These accounting boundaries are particularly relevant when interpreting differences between core and peripheral cities. Cities whose manufacturing sectors rely on emissions-intensive process industries, or whose electricity is supplied from outside the region, may have carbon footprints that differ from the energy-based values recorded here. The direction and magnitude of the resulting differences cannot be established without fuel-specific consumption data and process-emission inventories.

3.2.2 Digital economy development

Digital economy development was measured using a composite index adapted from the city-level measurement framework discussed by Zhao et al. [47].

Five indicators were included, following the city-level framework of Zhao et al. [47]: the digital financial inclusion index, the internet penetration rate (internet users per 100 residents), the share of employment in information

transmission, software, and information technology services, internet-related output per capita (telecommunications business volume per capita), and mobile internet users per 100 residents.

These indicators represented different dimensions of digital economy development. Digital financial inclusion, measured by the aggregate Peking University Digital Financial Inclusion Index, reflected access to financial services supported by digital technologies. The internet penetration rate and mobile internet subscriptions captured connectivity and individual adoption. The employment share of information technology services represented the supply of digital services and related human resources, while internet-related output per capita captured the economic scale of digital activity.

The five indicators were combined using the entropy-weighting method. For each year, the normalized value of indicator j for city i can be written as:

$$Z_{ijt} = \frac{x_{ijt} - \min_i (x_{ijt})}{\max_i (x_{ijt}) - \min_i (x_{ijt})} \tag{4}$$

where, x_{ijt} represents the original indicator value. The normalized indicator proportions were calculated as:

$$p_{ijt} = \frac{Z_{ijt}}{\sum_{i=1}^{N} Z_{ijt}} \tag{5}$$

The entropy value of each indicator was then calculated as:

$$e_{jt} = -\frac{1}{\ln N} \sum_{i=1}^{N} p_{ijt} \ln p_{ijt} \tag{6}$$

where, $N = 9$ denotes the number of cities. When $p_{ijt} = 0$, the corresponding term $p_{ijt} \ln p_{ijt}$ is defined as zero. Zero normalized values were replaced by 10^{-5} before the proportions were computed, so that the logarithmic terms remained defined; this replacement does not materially affect the resulting weights.

The entropy weight was obtained from:

$$w_{jt} = \frac{1 - e_{jt}}{\sum_{j=1}^{5} (1 - e_{jt})} \tag{7}$$

Finally, the composite digital economy index was constructed as:

$$DE_{it} = \sum_{j=1}^{5} w_{jt} Z_{ijt} \tag{8}$$

Entropy weights were calculated separately for each year, allowing the contributions of the five indicators to vary over time. This approach captured changes in the cross-sectional distribution of digital economy indicators but introduced a limitation for longitudinal comparisons because the normalization ranges and weights were not fixed across years.

The source values of the index for 2024 exhibited a documented standardization anomaly: values for all cities shifted to an implausibly low range relative to 2023. The 2024 index values used in this study were therefore recalculated with the same annual entropy-weighting procedure applied to the full national city sample, using the nine cities' 2024 indicator values compiled from municipal statistical sources. A specification excluding 2024 is reported as a robustness check in Section 4.9.

Accordingly, the index primarily represented the relative position of each city within the annual regional distribution. This characteristic was considered when interpreting the estimated relationship between digital economy development and manufacturing carbon emissions.

3.2.3 Mechanism and control variables

Two variables were used to examine the potential mechanisms linking digital economy development to manufacturing carbon emissions.

IND was measured as the share of secondary-industry value added in regional gross domestic product, expressed as a percentage. This variable served as a proxy for broad industrial restructuring. It did not directly measure changes in the composition of manufacturing subsectors.

R&D expenditure (RD) was used to represent innovation inputs. The natural logarithm of R&D expenditure, denoted by ln RD, was employed in the econometric analysis.

The baseline models also included economic development, foreign direct investment, energy intensity, IND, R&D expenditure, and precipitation as control variables. Per capita GDP and foreign direct investment were expressed in

logarithmic form. Energy intensity was measured using the percentage change in energy consumption per unit of output, while annual precipitation was included in logarithmic form.

IND and R&D expenditure were included as controls in the full baseline specifications and were examined separately as potential mediating variables in the mechanism analysis. The definitions and measurements of the study variables are presented in Table 2.

Table 2. Variable definitions and measurements

Variable	Symbol	Definition and Measurement
Manufacturing carbon emissions	$\ln \mathrm{CO}_2$	Natural logarithm of estimated manufacturing CO_2 emissions (10^4 t)
Carbon emission intensity	CI	CO_2 emissions per CNY 10,000 of manufacturing output (t)
Digital economy development	DE	Composite entropy-weighted index based on five digital economy indicators
Industrial structure	IND	Secondary-industry value added as a percentage of regional GDP
R&D expenditure	$\ln \mathrm{RD}$	Natural logarithm of research and development expenditure
Economic development	$\ln \mathrm{PGDP}$	Natural logarithm of per capita GDP
Foreign direct investment	$\ln \mathrm{FDI}$	Natural logarithm of utilized foreign direct investment
Energy intensity change	EI	Percentage change in energy consumption per unit of output
Annual precipitation	$\ln \mathrm{RA}$	Natural logarithm of annual precipitation

Note: CNY = Chinese yuan; R&D = research and development. The digital economy index was constructed using annually calculated entropy weights. Monetary variables were expressed in constant 2011 prices; emissions are measured in 10^4 tonnes of CO_2 (carbon-mass inventory converted at 44/12).

3.3 Baseline Econometric Model

A two-way fixed-effects model was first estimated to examine the association between digital economy development and manufacturing carbon emissions without explicitly incorporating spatial interactions.

The baseline specification was:

$$\ln \mathrm{CO}_{2,it} = \alpha + \beta \mathrm{DE}_{it} + \boldsymbol{\gamma}^{\mathsf{T}} \boldsymbol{X}_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{9}$$

where, i and t denote city and year, respectively; DE_{it} is the digital economy index; and $\boldsymbol{X}_{it}$ represents the vector of control variables. The parameter β measures the conditional association between digital economy development and manufacturing carbon emissions. The vector $\boldsymbol{\gamma}$ contains the coefficients of the control variables.

City fixed effects (μ_i) account for time-invariant characteristics of individual cities, while year fixed effects (λ_t) capture common temporal shocks. The error term is denoted by ε_{it}.

The fixed-effects specification controls for unobserved city characteristics that remain constant over time and for shocks common to all cities in a given year. It does not, however, eliminate potential bias arising from time-varying omitted variables or reverse causality. The estimated coefficients were therefore interpreted as conditional associations rather than causal effects.

3.4 Spatial Weight Matrices and Spatial Econometric Models

3.4.1 Construction of spatial weight matrices

Spatial dependence was considered because manufacturing activities in the PRD are connected through regional supply chains, infrastructure, labour markets, and industrial investment.

Two spatial weight matrices were specified to examine the sensitivity of the results to alternative representations of geographical relationships.

The first matrix, W_1, was a first-order contiguity matrix. Cities sharing a common boundary or vertex were assigned a spatial connection, while other city pairs were assigned zero:

$$w_{ij}^{(1)} = \begin{cases} 1, & \text{if cities } i \text{ and } j \text{ are contiguous,} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The diagonal elements were set to zero.

The second matrix, W_2, was based on inverse geographical distance:

$$w_{ij}^{(2)} = \begin{cases} \dfrac{1}{d_{ij}}, & 0 < d_{ij} \le d_0, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

where, d_{ij} represents the great-circle (Haversine) distance in kilometres between the seats of the municipal governments of cities i and j. No distance threshold was imposed: every city pair with $i \neq j$ receives the positive weight $1/d_{ij}$, so that no city is isolated in W_2. Coordinates refer to the seats of the municipal governments.

Both matrices were row-standardized:

$$\widetilde{w}_{ij} = \frac{w_{ij}}{\sum_{j=1}^{N} w_{ij}} \tag{12}$$

The row-standardized weights are denoted by w_{ij} in the subsequent equations. This normalization allows the spatially weighted value of a variable to be interpreted as a weighted average of neighbouring observations, provided that each row has a positive sum.

The contiguity matrix was used in the baseline spatial analysis, while the inverse-distance matrix was used as an alternative specification.

3.4.2 Spatial Durbin model

Three spatial econometric specifications were considered: the spatial autoregressive model (SAR), the spatial error model (SEM), and the spatial Durbin model (SDM).

The SDM incorporates spatial dependence in the dependent variable and spatially lagged explanatory variables. Its specification was:

$$\ln \mathrm{CO}_{2,it} = \rho \sum_{j=1}^{N} w_{ij} \ln \mathrm{CO}_{2,jt} + \beta \mathrm{DE}_{it} + \theta \sum_{j=1}^{N} w_{ij} \mathrm{DE}_{jt} + \boldsymbol{\gamma}^{\mathsf{T}} \boldsymbol{X}_{it} + \boldsymbol{\delta}^{\mathsf{T}} \sum_{j=1}^{N} w_{ij} \boldsymbol{X}_{jt} + \mu_i + \lambda_t + \varepsilon_{it} \tag{13}$$

where, ρ is the spatial autoregressive coefficient associated with neighbouring cities' manufacturing emissions; θ measures the coefficient of spatially lagged digital economy development; and $\boldsymbol{\delta}$ represents the coefficients of the spatially lagged control variables.

The spatial models were estimated by maximum likelihood. City and year fixed effects were entered as dummy variables, so that the estimates are equivalent to a least-squares dummy variable treatment of the fixed effects. All models used the full panel of 126 city–year observations. The standard error of the spatial autoregressive parameter was obtained from the numerically evaluated Hessian at the optimum, and the reported R^2 is an overall goodness-of-fit measure that includes the variation absorbed by the fixed effects. Standard errors are conventional throughout: no heteroskedasticity-robust or cluster-robust correction was applied to the reported estimates, and a nine-cluster check is reported in the note to Section 4.3.

The choice among spatial models was assessed using Lagrange multiplier (LM) tests, robust LM tests, likelihood comparisons, and likelihood ratio (LR) tests. The Hausman test was used to assess the suitability of fixed effects relative to random effects.

3.4.3 Decomposition of spatial effects

Following the spatial impact framework introduced in [37], the estimated association between digital economy development and manufacturing carbon emissions was decomposed into direct, indirect, and total effects.

In matrix notation, the SDM can be written as:

$$\boldsymbol{y}_t = \rho \boldsymbol{W} \boldsymbol{y}_t + \beta \mathrm{DE}_t + \theta \boldsymbol{W} \mathrm{DE}_t + \boldsymbol{X}_t \boldsymbol{\gamma} + \boldsymbol{W} \boldsymbol{X}_t \boldsymbol{\delta} + \boldsymbol{\mu} + \lambda_t \mathbf{1} + \boldsymbol{\varepsilon}_t \tag{14}$$

Rearranging the equation gives:

$$\boldsymbol{y}_t = (\boldsymbol{I} - \rho \boldsymbol{W})^{-1} \left(\beta \mathrm{DE}_t + \theta \boldsymbol{W} \mathrm{DE}_t + \boldsymbol{X}_t \boldsymbol{\gamma} + \boldsymbol{W} \boldsymbol{X}_t \boldsymbol{\delta} + \boldsymbol{\mu} + \lambda_t \mathbf{1} + \boldsymbol{\varepsilon}_t \right) \tag{15}$$

The partial derivative matrix associated with digital economy development is:

$$S_{\mathrm{DE}}(\boldsymbol{W}) = (\boldsymbol{I} - \rho \boldsymbol{W})^{-1} (\beta \boldsymbol{I} + \theta \boldsymbol{W}) \tag{16}$$

The average direct effect is obtained from the mean of the diagonal elements:

$$\mathrm{ADE} = \frac{1}{N} \operatorname{tr} \left[S_{\mathrm{DE}}(\boldsymbol{W}) \right] \tag{17}$$

The average total effect is calculated as:

$$\mathrm{ATE} = \frac{1}{N}\mathbf{1}^{\mathsf{T}} S_{\mathrm{DE}}(\boldsymbol{W})\mathbf{1} \tag{18}$$

The average indirect effect is then:

$$\mathrm{AIE} = \mathrm{ATE} - \mathrm{ADE} \tag{19}$$

The direct effect represents the average association between a city's digital economy development and its own manufacturing emissions, including feedback transmitted through the spatial system. The indirect effect captures the corresponding association with emissions in other cities, while the total effect combines both components.

Impact estimates are parameter point estimates evaluated at the reported SDM estimates using the partial-derivative expressions above. Their standard errors were obtained from 500 bootstrap resamples in which cities were resampled with replacement and the model was re-estimated for each resample.

3.5 Mechanism Analysis

To examine whether industrial restructuring and R&D expenditure served as potential channels linking digital economy development to manufacturing carbon emissions, a mediation regression framework was employed. The analysis followed the regression-based approach discussed in [43], with the significance of the estimated indirect associations assessed using the Sobel test.

Two mediating variables were considered separately: IND, measured by the share of secondary-industry value added in regional GDP, and R&D expenditure ($\ln \mathrm{RD}$), measured as the natural logarithm of research expenditure.

The mediation framework consisted of three regression equations:

$$Y_{it} = c\mathrm{DE}_{it} + \boldsymbol{\gamma}_1^{\mathsf{T}}\boldsymbol{Z}_{it} + \mu_i + \lambda_t + \varepsilon_{1,it} \tag{20}$$

$$M_{it} = a\mathrm{DE}_{it} + \boldsymbol{\gamma}_2^{\mathsf{T}}\boldsymbol{Z}_{it} + \mu_i + \lambda_t + \varepsilon_{2,it} \tag{21}$$

$$Y_{it} = c'\mathrm{DE}_{it} + bM_{it} + \boldsymbol{\gamma}_3^{\mathsf{T}}\boldsymbol{Z}_{it} + \mu_i + \lambda_t + \varepsilon_{3,it} \tag{22}$$

where, Y_{it} denotes the natural logarithm of manufacturing carbon emissions, DE_{it} represents digital economy development, and M_{it} is the mediating variable. The vector $\boldsymbol{Z}_{it}$ contains the additional controls included in the corresponding regression, excluding the mediating variable under examination. All three equations of each system included the same additional controls—$\ln \mathrm{PGDP}$, $\ln \mathrm{FDI}$, EI, and $\ln \mathrm{RA}$—together with city and year fixed effects, and were estimated on the same 126 observations. The two mediators were examined in separate systems and were never included simultaneously, so that the coefficients c, a, b, and c' of each system refer to an identical sample and control set. City and year fixed effects are represented by μ_i and λ_t, respectively.

The coefficient c represents the total association between digital economy development and manufacturing emissions in the specified non-spatial regression. The coefficient a measures the association between digital economy development and the mediator, while b represents the conditional association between the mediator and manufacturing emissions. The coefficient c' captures the association between digital economy development and emissions after accounting for the mediator.

The estimated indirect association was calculated as:

$$\mathrm{IE} = a \times b \tag{23}$$

Under a consistent linear mediation specification, the indirect association can also be expressed as:

$$\mathrm{IE} = c - c' \tag{24}$$

The Sobel statistic was calculated using:

$$z_{\mathrm{Sobel}} = \frac{ab}{\sqrt{b^2\mathrm{SE}_a^2 + a^2\mathrm{SE}_b^2}} \tag{25}$$

where, SE_a and SE_b denote the estimated standard errors of coefficients a and b, respectively.

The relative contribution of the indirect association was calculated as:

$$\mathrm{PM} = \frac{ab}{c} \times 100\% \tag{26}$$

where, PM is the estimated proportion mediated and c is the total coefficient from the corresponding non-spatial mediation model.

The industrial restructuring and R&D channels were examined independently. For industrial restructuring, $M_{it} = \mathrm{IND}_{it}$; for the innovation channel, $M_{it} = \ln \mathrm{RD}_{it}$. The estimated indirect associations were interpreted alongside their statistical significance rather than solely on the basis of the proportion mediated.

Because the analysis used observational city-level panel data, the mediation estimates were interpreted as statistical decompositions of conditional associations. They did not establish causal mediation in the absence of stronger identification assumptions. In addition, the mediation regressions did not incorporate the spatial feedback structure of the SDM. Their estimated indirect associations were therefore distinct from the spatial indirect effects discussed in Section 3.4.3.

3.6 Decoupling Analysis

The relationship between manufacturing output growth and carbon emissions was examined using the decoupling elasticity framework developed by Tapio [19].

The decoupling elasticity was defined as the ratio of the relative change in manufacturing carbon emissions to the relative change in manufacturing output:

$$T_{it} = \frac{(\mathrm{CO}_{2,it} - \mathrm{CO}_{2,i,t-1})/\mathrm{CO}_{2,i,t-1}}{(\mathrm{GM}_{it} - \mathrm{GM}_{i,t-1})/\mathrm{GM}_{i,t-1}} \tag{27}$$

where, T_{it} denotes the decoupling elasticity for city i in year t, $\mathrm{CO}_{2,it}$ represents manufacturing carbon emissions, and GM_{it} denotes manufacturing output.

The numerator measures the annual rate of change in manufacturing carbon emissions, while the denominator measures the corresponding rate of change in manufacturing output. Manufacturing output was used instead of aggregate GDP to maintain consistency with the sectoral focus of the study.

The decoupling states were classified according to the signs of the two growth rates and the value of the elasticity. Following the conventional Tapio classification, eight states were distinguished. The eight decoupling states and their corresponding classification criteria are summarized in Table 3.

Table 3. Classification of manufacturing carbon emission decoupling states

Decoupling State	Emission Growth	Output Growth	Elasticity
Strong decoupling	Negative	Positive	$T < 0$
Weak decoupling	Positive	Positive	$0 \leq T < 0.8$
Recessive decoupling	Negative	Negative	$T > 1.2$
Expansive coupling	Positive	Positive	$0.8 \leq T \leq 1.2$
Recessive coupling	Negative	Negative	$0.8 \leq T \leq 1.2$
Strong negative decoupling	Positive	Negative	$T < 0$
Expansive negative decoupling	Positive	Positive	$T > 1.2$
Weak negative decoupling	Negative	Negative	$0 \leq T < 0.8$

Note: T = decoupling elasticity. Decoupling states were determined jointly by the signs of emission and output growth and the elasticity thresholds. The classifications were not assigned using elasticity values alone.

Strong decoupling occurs when manufacturing output increases while carbon emissions decline. Weak decoupling indicates that both output and emissions increase, but emissions grow more slowly than output. Recessive decoupling describes a situation in which both variables decline, with emissions decreasing proportionally faster than manufacturing output.

Conversely, negative decoupling describes conditions under which emission performance deteriorates relative to output changes. Coupling represents situations in which manufacturing output and emissions change at approximately similar proportional rates.

The elasticity was calculated for consecutive years, producing a maximum of 13 annual observations for each city over the 2011–2024 period.

When the annual growth rate of manufacturing output approaches zero, the denominator of Eq. (27) becomes very small and the resulting elasticity may be extremely large. Two transitions in the sample were affected: Zhongshan in 2013 ($T = -710.7$) and Zhaoqing in 2016 ($T = -47.2$). In addition, the estimated 2024 output value for Zhaoqing implies zero growth, so the 2024 elasticity for that city is undefined and is excluded from its summary statistics. Section 4.7 therefore reports both the mean and the median of the annual elasticities for every city, so that the summary does not depend on a single extreme observation.

The decoupling analysis was used to characterize the relationship between manufacturing growth and emissions. It did not, by itself, identify the causal contribution of digital economy development to changes in decoupling status.

3.7 Descriptive Statistics

Table 4 presents the descriptive statistics for the variables used in the empirical analysis. The dataset comprised 126 observations from nine cities over 14 years.

Table 4. Descriptive statistics of the study variables

Variable	Obs.	Mean	Std. Dev.	Min.	Max.
$\ln \mathrm{CO_2}$	126	9.3466	0.7619	8.1653	10.6514
Emissions (10^4 t $\mathrm{CO_2}$)	126	14905.5507	9976.6159	3516.9	42251.3
DE	126	0.3738	0.2064	0.0665	0.8919
$\ln$ PGDP	126	11.4615	0.4498	10.2991	12.2342
PGDP (CNY)	126	104398.19	43990.51	29704.68	205714.26
$\ln$ FDI	126	13.8894	1.1223	10.9762	16.6322
IND (%)	126	43.9057	9.1941	20.7844	60.1473
EI (%)	126	−3.0667	3.0584	−8.7000	10.2450
$\ln$ RD	126	14.0469	1.1953	11.5125	17.0154
$\ln$ RA	126	7.5462	0.1675	7.0824	7.8845
CI (tonnes/CNY 10,000)	126	0.3522	0.1017	0.2110	0.7011

Note: CI = carbon emission intensity; CNY = Chinese yuan; DE = digital economy development; EI = energy intensity change; FDI = foreign direct investment; IND = industrial structure; ln = natural logarithm; Obs. = number of observations; PGDP = per capita gross domestic product; RA = annual precipitation; R&D = research and development; RD = R&D expenditure; Std. Dev. = standard deviation. Monetary variables are expressed in constant 2011 prices. Emissions are measured in 10^4 tonnes of $\mathrm{CO_2}$, obtained from the carbon-mass inventory through the 44/12 conversion; $\ln \mathrm{CO_2}$ denotes the natural logarithm of this emissions series.

The descriptive statistics show considerable variation in manufacturing carbon emissions and digital economy development across the nine cities. Manufacturing emissions averaged 14,905.55 $\times 10^4$ tonnes of $\mathrm{CO_2}$ with a standard deviation of 9,976.62, and ranged from 3,516.90 to 42,251.26 $\times 10^4$ tonnes. This dispersion indicates marked differences in the scale of manufacturing-related emissions across the study area.

The digital economy index had a mean of 0.3738 and a standard deviation of 0.2064, ranging from 0.0665 to 0.8919. This variation reflects differences in the annually weighted digital indicators among the nine cities. The group means reported below refer to the same panel: the weighted average of the core and peripheral group means (0.5016 and 0.2715) reproduces the overall mean of 0.3738.

The IND variable averaged 43.91%, indicating that secondary-industry activities accounted for a substantial proportion of regional economic output. The mean annual change in energy intensity was −3.07%, although the observed values varied considerably across cities and years.

Differences between core and peripheral cities were also apparent in the reported subgroup statistics. Guangzhou, Shenzhen, Foshan, and Dongguan had a mean digital economy index of 0.502, compared with 0.272 for Zhuhai, Huizhou, Zhongshan, Jiangmen, and Zhaoqing. The corresponding mean logarithmic R&D expenditure values were 15.074 and 13.225.

The peripheral cities had a slightly higher average secondary-industry share, at 45.06%, compared with 42.47% in the core cities.

Taken together, the descriptive results indicate substantial differences in digital development, industrial composition, and manufacturing emissions within the PRD. These differences provide a basis for the subsequent examination of regional heterogeneity.

4 Empirical Results

4.1 Spatial Dependence in Manufacturing Carbon Emissions

Spatial autocorrelation was examined before estimating the spatial panel models. Global Moran's I statistics were calculated for manufacturing carbon emissions and the digital economy index in 2011, 2015, 2019, and 2024 using the row-standardized contiguity matrix (W_1). Statistical significance was assessed using both the normal approximation and permutation tests based on 999 random permutations. The results are reported in Table 5.

The reported Moran's I statistics for manufacturing carbon emissions increased from 0.0130 in 2011 to 0.4647 in 2024. The 2011 estimate was not statistically significant, whereas the 2015 estimate was marginally significant under the normal approximation ($p = 0.0696$), but not under the permutation test ($p = 0.1051$). By 2019, the statistic had increased to 0.3735 and was significant under both testing procedures. The corresponding value reached 0.4647 in 2024, with both p-values below 0.01.

These results indicate that the spatial clustering of manufacturing emissions became more pronounced across the four reported years. Cities with relatively high emissions were increasingly located near other high-emission

cities, while lower-emission cities tended to be located near one another. Such patterns are consistent with the spatial concentration of manufacturing activities within the regional production system.

Table 5. Global Moran's I statistics for manufacturing carbon emissions and digital economy development

Year	Variable	Moran's I	z-statistic	Normal p-Value	Permutation p-Value
2011	$\ln CO_2$	0.0130	0.630	0.5286	0.9560
2011	DE	0.1229	1.146	0.2519	0.5095
2015	$\ln CO_2$	0.2744	1.815	0.0696	0.1051
2015	DE	−0.0122	0.518	0.6047	0.9459
2019	$\ln CO_2$	0.3735	2.258	0.0239	0.0180
2019	DE	−0.1535	−0.134	0.8932	0.4054
2024	$\ln CO_2$	0.4647	2.671	0.0076	0.0050
2024	DE	0.0253	0.686	0.4925	0.8929

Note: DE = digital economy development. $\ln CO_2$ denotes the natural logarithm of this emissions series. W_1 is the row-standardized first-order contiguity matrix. Permutation p-values were obtained from 999 random permutations.

The results do not, however, establish that industrial relocation caused the observed clustering. Similar spatial patterns may also arise from differences in industrial specialization, energy consumption, infrastructure, or other geographically concentrated economic characteristics.

In contrast, the digital economy index showed no statistically significant spatial autocorrelation in any of the four years. Its Moran's I values remained relatively small and changed sign over time. The difference between the spatial distributions of digital development and manufacturing emissions suggests that the two variables did not exhibit identical geographical patterns.

These findings provide a descriptive basis for examining spatial dependence in manufacturing emissions. The suitability of the spatial econometric specifications was assessed further using formal model diagnostic tests.

4.2 Spatial Model Selection

The spatial econometric specifications were compared using LM tests, robust LM tests, LR tests, and the Hausman test. The reported results are presented in Table 6.

Table 6. Spatial dependence diagnostics and model specification tests

Test	Statistic	Degrees of Freedom	p-Value
LM-Lag	0.186	1	0.667
LM-Error	1.177	1	0.278
Robust LM-Lag	0.143	1	0.705
Robust LM-Error	60.157	1	<0.001
LR: SDM versus SAR	19.110	7	0.008
LR: SDM versus SEM	24.628	7	0.001
Hausman test	27.050	7	<0.001

Note: LM = Lagrange multiplier; LR = likelihood ratio; SAR = spatial autoregressive model; SDM = spatial Durbin model; SEM = spatial error model. The reported specifications included city and year fixed effects.

The conventional LM tests did not reject the null hypotheses of no spatial lag dependence or no spatial error dependence. The LM-Lag statistic was 0.186 (p = 0.667), while the LM-Error statistic was 1.177 (p = 0.278). The robust LM-Lag test was also insignificant. By contrast, the robust LM-Error statistic was 60.157 ($p < 0.001$), indicating evidence of spatial dependence under that diagnostic specification.

The LR tests provided further information on model selection. The reported LR comparisons favoured the SDM over its SAR and SEM restrictions. The statistic was 19.110 for the SAR restriction (reported p = 0.008) and 24.628 for the SEM restriction (reported p = 0.001). These comparisons assume identical samples and consistent likelihood definitions.

The Hausman test statistic was 27.050 ($p < 0.001$), favouring the fixed-effects specification over the random-effects alternative under the reported test assumptions.

The two-way fixed-effects SDM was selected on the basis of the likelihood-ratio comparisons.

4.3 Baseline Regression Results

Table 7 reports the estimates from two-way fixed-effects models and the three spatial panel specifications. The reduced fixed-effects model excluded IND and R&D expenditure, while the full specification included the complete

set of control variables.

Table 7. Baseline fixed-effects and spatial panel regression results

Variable	FE (Reduced)	FE (Full)	SAR	SEM	SDM
DE	−0.6351***	−0.4313**	−0.3583**	−0.3434**	−0.3423**
	(0.1825)	(0.1723)	(0.1476)	(0.1520)	(0.1631)
ln PGDP	0.7334***	0.5373***	0.4672***	0.4813***	0.3725***
	(0.0909)	(0.1053)	(0.0902)	(0.0941)	(0.1069)
ln FDI	0.0788***	0.0647**	0.0498**	0.0515**	0.0230
	(0.0279)	(0.0263)	(0.0225)	(0.0232)	(0.0267)
IND	—	0.0157***	0.0184***	0.0160***	0.0193***
		(0.0045)	(0.0039)	(0.0039)	(0.0038)
EI	0.0053	0.0011	0.0006	0.0010	0.0029
	(0.0060)	(0.0056)	(0.0048)	(0.0049)	(0.0046)
ln RD	—	0.1159*	0.1167**	0.1153**	0.0573
		(0.0644)	(0.0551)	(0.0559)	(0.0560)
ln RA	−0.1020	−0.0750	−0.0514	−0.0743	0.0501
	(0.1571)	(0.1440)	(0.1233)	(0.1250)	(0.1223)
$W \times$ DE	—	—	—	—	0.2263*
					(0.1230)
$W \times$ ln PGDP	—	—	—	—	0.1461
					(0.1026)
$W \times$ ln FDI	—	—	—	—	0.0598*
					(0.0309)
$W \times$ IND	—	—	—	—	0.0027
					(0.0033)
$W \times$ EI	—	—	—	—	0.0202***
					(0.0070)
$W \times$ ln RD	—	—	—	—	−0.0755**
					(0.0317)
$W \times$ ln RA	—	—	—	—	−0.0456
					(0.1188)
ρ/λ	—	—	0.0854***	−0.1003	0.1220*
			(0.0316)	(0.0830)	(0.0635)
Log-likelihood	—	—	100.91	98.15	110.46
Overall R^2	0.9734	0.9783	0.9796	0.9787	0.9825

Note: DE = digital economy development; EI = energy intensity change; FDI = foreign direct investment; FE = fixed effects; IND = industrial structure; PGDP = per capita gross domestic product; RA = annual precipitation; RD = research and development expenditure; SAR = spatial autoregressive model; SDM = spatial Durbin model; SEM = spatial error model; ρ = spatial autoregressive parameter; λ = spatial error autoregressive parameter. Conventional (non-robust) standard errors from the least-squares dummy variable (FE) or maximum-likelihood (SAR, SEM, SDM) estimates are reported in parentheses, and tests use standard normal or χ^2 reference distributions. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$. The spatial models included the full set of controls. W denotes the spatial weight matrix; $W\times$ indicates a spatially lagged regressor, included in the SDM only. The reported R^2 is an overall measure including the variation absorbed by city and year fixed effects. As a sensitivity check, the city-cluster-robust standard error of the DE coefficient in the FE (full) specification, with nine clusters, is 0.2198 (t = −1.96). "—" indicates not applicable.

The digital economy coefficient was negative across all five specifications. In the reduced fixed-effects model, the coefficient was −0.6351 and was statistically significant at the 1% level. Its magnitude decreased to −0.4313 after IND and R&D expenditure were included, although the coefficient remained significant at the 5% level.

The spatial specifications produced smaller negative coefficients. The estimates were −0.3583 for the SAR model, −0.3434 for the SEM model, and −0.3423 for the SDM. Each coefficient was reported as statistically significant at the 5% level.

The baseline SDM coefficient for DE was −0.3423 (SE 0.1631). Because the model includes spatial feedback, its magnitude is interpreted alongside the direct, indirect, and total impacts presented in Section 4.4.

The estimated coefficient of the spatially lagged dependent variable was positive in the SDM ($\rho = 0.1220$, $p < 0.10$). This suggests a positive conditional spatial relationship in manufacturing emissions under the specified weight matrix. It does not, by itself, establish a significant spatial spillover from digital economy development.

The spatially lagged coefficients are reported in the lower section of Table 7. The coefficient on W × DE was positive (0.2263, SE 0.1230, p = 0.066) and that on W × EI was positive and significant (0.0202, SE 0.0070, p = 0.004), whereas the coefficient on W × $\ln$ RD was negative (-0.0755, SE 0.0317, p = 0.017). These coefficients describe conditional associations between neighbouring cities' characteristics and local emissions.

Among the control variables, per capita GDP was positively associated with manufacturing emissions across all specifications. In the SDM, the estimated coefficient was 0.3725 and was statistically significant at the 1% level. The IND coefficient was also positive and significant, with an SDM estimate of 0.0193. The positive coefficient describes a conditional parameter association; its average marginal interpretation in the SDM requires the spatial impact decomposition.

Foreign direct investment was positively associated with emissions in the non-spatial models, SAR, and SEM, but its coefficient was not statistically significant in the SDM. The R&D expenditure coefficient was positive and significant in several specifications but became insignificant in the SDM. Neither the energy intensity change variable nor annual precipitation was statistically significant in the reported baseline regressions.

Overall, the negative coefficient for digital economy development was consistent across the reported model specifications. Nevertheless, the statistical interpretation of this relationship depends on the spatial impact estimates and the assumptions underlying the model.

4.4 Spatial Impact Decomposition

The coefficients of a SDM do not directly represent the marginal effects of changes in explanatory variables. Spatial feedback may transmit an initial change in one city to neighbouring cities and subsequently back to the originating city. Accordingly, the estimated SDM was decomposed into average direct, indirect, and total effects using the framework described in [37]. The estimated direct, indirect, and total effects are reported in Table 8.

Table 8. Direct, indirect, and total effects from the spatial Durbin model

Variable	Direct Effect	Indirect Effect	Total Effect
DE	−0.3346	0.2026	−0.1320
	(0.3191)	(0.2982)	(0.5234)
$\ln$ PGDP	0.3805	0.2102	0.5907*
	(0.2796)	(0.2060)	(0.3013)
$\ln$ FDI	0.0256	0.0687	0.0942*
	(0.0313)	(0.0437)	(0.0553)
IND	0.0195	0.0056	0.0251*
	(0.0122)	(0.0056)	(0.0141)
EI	0.0037	0.0225*	0.0263*
	(0.0054)	(0.0131)	(0.0154)
$\ln$ RD	0.0545	−0.0752	−0.0207
	(0.1196)	(0.0904)	(0.1587)
$\ln$ RA	0.0485	−0.0434	0.0051
	(0.1807)	(0.2063)	(0.3109)

Note: DE = digital economy development; EI = energy intensity change; FDI = foreign direct investment; IND = industrial structure; ln = natural logarithm; PGDP = per capita gross domestic product; RA = annual precipitation; RD = research and development expenditure; SDM = spatial Durbin model. Effects are parameter point estimates evaluated at the SDM estimates reported in Table 7, using the partial-derivative matrix expressions in Section 3.4.3. Standard errors are shown in parentheses and were obtained from 500 bootstrap resamples in which cities were resampled with replacement. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$. Under the row-standardized W_1, the total effect of a regressor equals $(\beta + \theta)/(1 - \rho)$; the reported total effects satisfy this identity exactly.

The estimated average direct effect of digital economy development was -0.3346, the average indirect effect was 0.2026, and the average total effect was -0.1320. The direct-effect point estimate was close to the SDM coefficient of -0.3423, and the feedback correction was small and positive, as implied by a positive value of $\theta + \rho\beta$. The indirect effect was positive but imprecisely estimated. The point estimates are consistent with the parameter-based identity for a row-standardized weight matrix, $(\beta+\theta)/(1-\rho) = -0.1320$, which the reported total effect reproduces.

This result provided no statistically significant evidence of cross-city spillovers associated with digital economy development under the specified model. The positive indirect-effect point estimate of 0.2026 was imprecisely estimated and did not establish that digitalization displaced emissions to neighbouring cities; neither did the estimate support a significant negative spillover associated with the diffusion of cleaner production technologies.

An important distinction emerged between the SDM regression coefficient and the impact decomposition. Although the coefficient of DE in Table 7 was significant at the 5% level, the bootstrap standard errors of the

decomposed direct and total effects were 0.3191 and 0.5234, respectively. Neither effect was statistically significant at conventional levels.

The decomposition therefore provided less precise evidence concerning the marginal effects of digital economy development than the coefficient estimates alone might suggest. This distinction is important when evaluating H1 and H4. The baseline coefficient was consistent with H1, but the decomposed direct and total effects did not provide statistically significant confirmation of the corresponding spatial marginal effects. The estimated indirect effect did not provide statistically significant support for H4.

4.5 Industrial Structure as a Statistical Channel

Table 9 reports the non-spatial industrial-structure decomposition. The DE coefficient in the mediator equation was −10.2465, and the conditional IND coefficient in the emissions equation was 0.0185. Their product was approximately −0.1894. The reported Sobel statistic was −2.234 (p = 0.026), with a reported mediated share of 29.82%. These quantities refer to the non-spatial total association of −0.6351, not to the SDM coefficient of −0.3423. The direct association remained negative after adding IND. The result is consistent with H2 when industrial composition is measured by the secondary-industry share. This measure does not distinguish changes within manufacturing from broader shifts between secondary and tertiary activities.

Table 9. Industrial-structure mediation regressions

Equation	Outcome	DE Coefficient (SE)	IND Coefficient (SE)
Total association	$\ln CO_2$	−0.6351 (0.1825)	—
Mediator	IND	−10.2465 (3.9182)	—
Conditional association	$\ln CO_2$	−0.4458 (0.1741)	0.0185 (0.0043)

Note: DE = digital economy development; SE = standard error; EI = energy intensity change; IND = industrial structure; FDI = foreign direct investment; ln = natural logarithm; PGDP = per capita gross domestic product; RA = annual precipitation; RD = research and development expenditure. All three equations were estimated on the same 126 observations with city and year fixed effects and the additional controls $\ln$ PGDP, $\ln$ FDI, EI, and $\ln$ RA. $\ln$ RD was not included in this system. Standard errors are conventional (non-robust) standard errors. The estimated indirect association was −0.1894; the Sobel statistic was −2.234 (p = 0.026), and the estimated proportion mediated was 29.82%. "—" indicates not applicable.

4.6 R&D Expenditure as a Statistical Channel

The R&D results in Table 10 did not support H3. The DE coefficient in the $\ln$ RD equation was −0.3685 (SE 0.2749), while $\ln$ RD was positively associated with emissions in the conditional equation. The reported product was −0.0707, with Sobel z = −1.224 and p = 0.221. Its arithmetic ratio to the non-spatial total coefficient was 11.13%, but that ratio is not evidence of a statistically established contribution. The finding concerns aggregate research expenditure; it does not test the environmental performance of individual innovations or rule out a role for adopted technologies.

Table 10. R&D mediation regressions

Equation	Outcome	DE Coefficient (SE)	$\ln$ RD Coefficient (SE)
Total association	$\ln CO_2$	−0.6351 (0.1825)	—
Mediator	$\ln$ RD	−0.3685 (0.2749)	—
Conditional association	$\ln CO_2$	−0.5644 (0.1772)	0.1919 (0.0639)

Note: R&D = research and development; DE = digital economy development; EI = energy intensity change; FDI = foreign direct investment; IND = industrial structure; ln = natural logarithm; PGDP = per capita gross domestic product; RA = annual precipitation; RD = research and development expenditure; SE = standard error. All three equations were estimated on the same 126 observations with city and year fixed effects and the additional controls $\ln$ PGDP, $\ln$ FDI, EI, and $\ln$ RA. IND was not included in this system. Standard errors are conventional (non-robust) standard errors. The estimated indirect association was −0.0707; the Sobel statistic was −1.224 (p = 0.221), and the estimated proportion mediated was 11.13%. "—" indicates not applicable.

4.7 Manufacturing Growth and Emissions Decoupling

Table 11 summarizes the city-level decoupling results. Weak decoupling occurred frequently, including 11 annual transitions in Shenzhen and 10 in Dongguan. Guangzhou recorded six strong-decoupling years and Foshan four. Weak decoupling was not a majority outcome in every city: Guangzhou recorded three weak-decoupling years

and Huizhou four out of a maximum of 13 annual transitions. Shenzhen's reported average elasticity of 0.440 was also higher than Guangzhou's 0.254 and Foshan's 0.268.

Table 11. Reported manufacturing decoupling summaries

City	Mean Elasticity	Median Elasticity	Strong Decoupling	Weak Decoupling	Expansive Negative Decoupling	Recessive Decoupling	Other States
Guangzhou	0.254	0.151	6	3	1	3	0
Shenzhen	0.440	0.408	0	11	0	0	2
Zhuhai	0.665	0.579	2	7	1	2	1
Foshan	0.268	0.344	4	6	0	1	2
Huizhou	0.355	0.695	2	4	3	0	4
Dongguan	0.492	0.528	2	10	1	0	0
Zhongshan	$-53.901^{\dagger}$	0.469	2	8	1	1	1
Jiangmen	1.003	0.693	2	6	1	2	2
Zhaoqing	$-2.553^{\dagger}$	0.524	2	7	1	1	2

Note: Counts denote annual transitions; each city has at most 13 transitions over 2011–2024. Mean and median summarize the annual elasticities of each city. Two transitions with |elasticity| > 20, Zhongshan 2013 ($T = -710.7$) and Zhaoqing 2016 ($T = -47.2$), dominate the corresponding means; † marks the two cities whose means are affected, and excluding the extreme transition yields means of 0.829 for Zhongshan and 1.509 for Zhaoqing. Medians are reported alongside means so that the summary of each city does not depend on a single observation. Zhaoqing's 2024 transition is undefined because the estimated output growth rate is zero, so its statistics use 12 transitions. The Other states column comprises expansive coupling (six transitions), recessive coupling (three), weak negative decoupling (two), and strong negative decoupling (two); the undefined Zhaoqing 2024 transition is included in the Zhaoqing Other states count.

The decoupling summaries describe the joint movement of manufacturing output and emissions. They are presented as supplementary descriptive evidence and are not used to estimate the effect of digital economy development on decoupling.

4.8 Regional Heterogeneity

To examine whether the relationship between digital economy development and manufacturing carbon emissions differed across the PRD, the sample was divided into two groups according to manufacturing scale measured at the beginning of the sample period. The four cities with the largest manufacturing output value in 2011—Shenzhen (19,110 $\times$ 10^8 CNY), Guangzhou (14,505), Foshan (13,952), and Dongguan (7,893)—were classified as core cities, and the remaining five—Zhongshan, Huizhou, Jiangmen, Zhuhai, and Zhaoqing—as peripheral cities. The grouping was fixed from the 2011 output ranking before the regression results were examined, and it coincides with the core–periphery distinction commonly used in studies of the PRD.

Separate regressions were estimated for the two groups. The results are presented in Table 12.

The subgroup regressions yielded coefficients with contrasting signs. In the core cities, digital economy development was negatively associated with manufacturing carbon emissions, with a coefficient of -0.9511 and a standard error of 0.1744. The coefficient was statistically significant at the 1% level. In contrast, the estimated coefficient for the peripheral cities was positive (0.3047) but statistically insignificant.

These findings suggested that the negative association observed in the full sample was more evident among the four core manufacturing cities. The difference may reflect variations in digital infrastructure, industrial specialization, production efficiency, and the maturity of digital technology adoption. Core cities generally have more developed digital services and a greater concentration of advanced manufacturing activities, whereas peripheral cities retain different INDs and production functions.

The estimated coefficients for IND were positive in both groups, although the point estimate was larger in the peripheral cities (0.0278) than in the core cities (0.0069). A pooled interaction model, in which each coefficient was interacted with a core-city indicator, was used to test the differences formally. For digital economy development the interaction coefficient was -0.9059 (SE 0.2935; $z = -3.087$, $p = 0.002$), confirming that the DE coefficient differs significantly between the two groups. The corresponding interaction coefficient for IND was -0.0077 (SE 0.0066; $z = -1.165$, $p = 0.244$), and that for R&D expenditure was -0.1062 (SE 0.0467; $z = -2.273$, $p = 0.023$).

R&D expenditure also showed different relationships with emissions. Its coefficient was negative but insignificant in the core cities (-0.0344), whereas the peripheral-city coefficient was positive and statistically significant (0.2635, $p < 0.05$). This difference suggested that R&D expenditure may have been associated with different industrial activities across the two groups. However, expenditure alone did not indicate whether innovation was directed towards cleaner production, capacity expansion, or other technological objectives.

Table 12. Regional heterogeneity in the relationship between digital economy development and manufacturing carbon emissions

Variable	Core Cities	Peripheral Cities
DE	−0.9511***	0.3047
	(0.1744)	(0.3263)
ln PGDP	0.6983***	0.2511
	(0.0961)	(0.1587)
ln FDI	0.0371	0.0086
	(0.0255)	(0.0448)
IND	0.0069*	0.0278***
	(0.0042)	(0.0090)
EI	−0.0050	0.0126
	(0.0074)	(0.0079)
ln RD	−0.0344	0.2635**
	(0.0515)	(0.1123)
ln RA	0.0895	−0.0937
	(0.1530)	(0.2147)
Observations	56	70

Note: DE = digital economy development; EI = energy intensity change; FDI = foreign direct investment; IND = industrial structure; ln = natural logarithm; PGDP = per capita gross domestic product; RA = annual precipitation; RD = research and development expenditure. Conventional (non-robust) standard errors are reported in parentheses. *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$. Core cities include Guangzhou, Shenzhen, Foshan, and Dongguan; peripheral cities include Zhuhai, Huizhou, Zhongshan, Jiangmen, and Zhaoqing. The subgroups contain 56 and 70 observations, respectively, from four and five cities; at this group size, cluster-robust inference is not reliable, and the subgroup estimates should be interpreted as descriptive.

Two explanations may account for the contrasting digital economy coefficients. First, the environmental implications of digital development may depend on the level of industrial digitalization and the ability of firms to apply digital technologies to energy management and production optimization. Second, differences in the location of emissions-intensive manufacturing activities may affect the observed relationship between digital development and local carbon emissions.

These explanations remain tentative because the available data did not directly measure intercity production transfers or the environmental performance of adopted technologies. Nevertheless, the subgroup estimates indicated that regional differences should be considered when interpreting the relationship between digital economy development and manufacturing decarbonization.

4.9 Robustness Checks

Several alternative specifications were considered to assess the sensitivity of the baseline findings: replacing the first-order contiguity matrix with an inverse-distance weight matrix, using the logarithm of manufacturing carbon intensity as an alternative dependent variable, and re-estimating the model on the subsample that excludes 2024, the year in which the source values of the digital economy index required recalculation.

The reported results are summarized in Table 13.

Table 13. Robustness checks

Model Specification	DE Coefficient	Standard Error	Test Statistic
Baseline SDM (W_1)	−0.3423	0.1631	−2.098
Inverse-distance SDM (W_2)	−0.3621	0.1528	−2.369
SDM with log carbon intensity (W_1)	0.1378	0.0937	1.470
SDM excluding 2024 (W_1)	−0.3699	0.1484	−2.493

Note: DE = digital economy development; SDM = spatial Durbin model; W_1 = row-standardized first-order contiguity matrix; W_2 = row-standardized inverse-distance matrix. Each row is a separate SDM estimated on the full panel of 126 observations with the same controls and city and year fixed effects, except for the last row, which uses the 117 observations from 2011 to 2023. Test statistics are z-statistics based on conventional standard errors. W_1 denotes the row-standardized contiguity matrix and W_2 the row-standardized inverse-distance matrix.

The alternative spatial weight matrix produced a negative coefficient for digital economy development (−0.3621), which was statistically significant at the 5% level. This estimate was similar in direction and magnitude to the baseline

SDM coefficient of -0.3423, suggesting that the negative coefficient was not confined to the first-order contiguity specification.

When the logarithm of manufacturing carbon intensity replaced total emissions as the dependent variable, the coefficient of digital economy development was positive and statistically insignificant (0.1378, SE 0.0937, $z = 1.47$). Because carbon intensity is the ratio of emissions to manufacturing output, this result indicates that the negative association with the level of emissions operates through the scale and composition of manufacturing activity rather than through emissions per unit of output. The outcome does not contradict the baseline estimate, but it qualifies its interpretation: digital development in the sample is associated with lower total emissions rather than with a lower emission content of each unit of output.

Re-estimating the SDM without the 2024 observations, whose source values for the digital economy index required recalculation, produced a coefficient of -0.3699 (SE 0.1484, $z = -2.49$). The baseline result therefore does not depend on the inclusion of 2024.

The alternative weight matrix and the exclusion of 2024 both yielded negative and statistically significant DE coefficients, whereas the carbon-intensity specification did not. These checks assess sensitivity to the spatial weighting scheme, the treatment of the 2024 index values, and the outcome definition; they do not address all sources of endogeneity.

5 Discussion

5.1 Digital Economy Development and Manufacturing Decarbonization

The empirical analysis revealed a negative relationship between digital economy development and manufacturing carbon emissions in the baseline fixed-effects and spatial panel specifications. The coefficient remained negative and significant when an alternative spatial weight matrix was used and when the 2024 observations were excluded; under the carbon-intensity specification it was positive and statistically insignificant.

These findings were broadly consistent with previous research reporting negative associations between digital development and carbon emission performance in China [1–3, 5]. However, the present study also showed that the relationship varied across cities within a relatively integrated manufacturing region.

The distinction between the baseline SDM coefficient and the spatial impact estimates is important. Although the coefficient for digital economy development in the SDM was statistically significant, the estimated average direct effect and total effect were not statistically significant at conventional levels. Accordingly, the results provide evidence of a negative conditional relationship in the reported regression specifications, but the spatial impact decomposition does not establish a statistically significant average marginal effect.

This distinction matters for the interpretation of digitalization as a low-carbon development strategy. Digital technologies may support more efficient production, better resource allocation, and changes in industrial composition. At the same time, digital infrastructure requires additional energy, and improvements in productivity may stimulate manufacturing output. The environmental outcome therefore depends on how digital technologies interact with local industrial activities.

The findings suggest that digital development should not be treated as a uniform source of manufacturing emission reductions. Its association with carbon performance appears to depend on the structure and development stage of the regional manufacturing economy.

5.2 Industrial Restructuring and the Role of Innovation

The mediation analysis found a statistically significant indirect association involving the secondary-industry share. The estimated indirect association accounted for 29.82% of the total association in the non-spatial mediation model.

This result was consistent with the theoretical argument that changes in industrial composition can affect regional carbon performance. Digital connectivity may facilitate the reorganization of production networks and the expansion of higher-value economic activities. Such changes can alter the relative importance of manufacturing and service activities within local economies.

However, the IND variable measured the share of secondary-industry value added in regional GDP. It did not distinguish between cleaner and more emissions-intensive manufacturing subsectors. Consequently, the mediation result cannot establish whether the observed relationship arose from improvements within manufacturing, service-sector expansion, or changes in the geographical distribution of production.

The R&D channel presented a different pattern. The estimated indirect association was negative but statistically insignificant. This finding did not support R&D expenditure as a statistically detectable mediating variable in the present sample.

One possible explanation concerns the distinction between innovation inputs and technological outcomes. R&D expenditure measures the resources allocated to research activities, but it does not directly capture the adoption of energy-saving technologies or the environmental performance of innovation. Research expenditure may also

accompany the expansion of industrial production, particularly in cities where manufacturing capacity and capital investment continue to grow.

The findings therefore point to the importance of distinguishing industrial restructuring from technological innovation when examining manufacturing decarbonization. They also suggest that future research should incorporate more direct indicators of green technological outcomes, including environmentally related patents, industrial energy efficiency, and the adoption of low-carbon production technologies.

5.3 Spatial Dependence and Regional Differences

The spatial analysis showed that manufacturing carbon emissions became more spatially clustered during the study period. The reported Moran's I increased from 0.0130 in 2011 to 0.4647 in 2024, with statistically significant positive spatial autocorrelation in the later years.

This pattern indicated that cities with similar manufacturing emission levels became more geographically concentrated. However, spatial clustering alone does not identify the processes responsible for this pattern. Industrial relocation, differences in manufacturing specialization, common regional shocks, and changes in energy use may all contribute to spatial dependence.

The spatial impact decomposition did not identify a statistically significant indirect effect of digital economy development on manufacturing emissions. The estimated indirect effect was positive (0.2026) with a bootstrap standard error of 0.2982, and the corresponding total effect was -0.1320 with a standard error of 0.5234. Neither estimate differs from zero at conventional levels.

This finding provided no clear statistical support for either negative emission spillovers associated with cleaner technology diffusion or positive spillovers associated with emissions displacement. The absence of a statistically significant indirect effect should not be interpreted as evidence that cross-city production relationships are unimportant. Rather, such relationships were not detected through the estimated spatial impact measure.

The regional subgroup analysis further indicated differences between core and peripheral cities. The DE coefficient was negative and statistically significant in the pooled core-city subsample and positive but statistically insignificant in the peripheral-city subsample.

These differences may reflect variations in industrial specialization, digital infrastructure, and the types of production activities undertaken in different cities. The contrasting subgroup estimates are consistent with differences in industrial and digital development conditions across cities.

The available evidence does not establish that the differences were caused by the relocation of emissions-intensive production. Nevertheless, the findings support the need to examine manufacturing decarbonization at both city and regional levels, particularly where industrial supply chains cross administrative boundaries.

5.4 Policy Implications

The findings have several implications for digital development and low-carbon manufacturing policy in the PRD and other manufacturing-oriented urban regions.

First, because fuel-by-fuel physical consumption for the manufacturing sector is not published at the city level, the fuel-level sum in Eq. (1) was evaluated at the standard-coal aggregate with the raw-coal coefficient, so the procedure does not identify the fuel mix, and industrial process emissions such as those from cement and steel production are not separately counted. The series is measured in tonnes of CO_2 after the 44/12 conversion. Future research could incorporate fuel-specific combustion data, electricity-related emissions, and process emissions to provide a more complete assessment.

Second, industrial restructuring policies should consider carbon performance across the wider regional production network. Changes in the industrial composition of individual cities may reduce locally recorded emissions without producing equivalent reductions across the region. Policies supporting industrial relocation should therefore incorporate energy-efficiency requirements, emissions reporting, and cleaner production standards for receiving industrial parks and enterprises.

Third, innovation policies should distinguish between research expenditure and environmental outcomes. R&D expenditure measures innovation inputs rather than environmental outcomes. Innovation programmes could therefore be evaluated using measured energy and carbon outcomes alongside research expenditure. Evaluation systems could place greater weight on improvements in industrial energy efficiency, the adoption of cleaner production processes, and measurable reductions in emissions per unit of output.

Fourth, regional coordination remains important even where statistical spillover effects are not detected. Shared emissions-monitoring arrangements, consistent industrial carbon accounting standards, and cooperation between core-city technology providers and peripheral manufacturing enterprises may improve the evaluation of regional decarbonization efforts.

These policy considerations are relevant to other developing manufacturing regions where digital infrastructure is expanding alongside changes in industrial composition. The central implication is that digital development and

low-carbon manufacturing policy should be coordinated with IND, technological capacity, and the geographical organization of production.

5.5 Limitations and Future Research

Several limitations should be considered when interpreting the findings.

First, manufacturing carbon emissions were estimated from the direct combustion of eight energy sources. The inventory did not include indirect emissions associated with purchased electricity or process-related emissions from industries such as cement and steel. These accounting boundaries may affect comparisons among cities with different energy systems and INDs. Future research could incorporate electricity-related emissions, industrial process emissions, and supply-chain carbon accounting to provide a more complete assessment of regional manufacturing carbon performance.

Second, the IND variable represented the share of secondary-industry value added in regional GDP. Although suitable for describing broad changes in economic composition, it did not directly measure restructuring within manufacturing. More detailed information on manufacturing subsectors, production relocation, and sector-specific carbon intensity would allow the industrial restructuring mechanism to be examined more precisely.

Third, R&D expenditure was an input-based indicator of innovation. It did not capture the direction, adoption, or environmental performance of technological change. Future studies could use green patent indicators, industrial energy-saving technology adoption, and firm-level innovation outcomes to distinguish general innovation activity from innovation directly related to manufacturing decarbonization.

Fourth, the digital economy index was constructed using annually calculated entropy weights. This approach captured differences in the annual distribution of digital indicators but limited strict comparisons of index levels across years. A composite index based on common normalization parameters and fixed weights would provide a useful alternative for examining longitudinal relationships.

Finally, the empirical models identified conditional associations rather than causal effects. City and year fixed effects accounted for time-invariant city characteristics and common temporal shocks, but time-varying confounding factors and reverse causality remained possible. Future studies could examine suitable policy interventions, externally driven infrastructure changes, or credible instrumental-variable strategies to strengthen causal identification.

6 Conclusions

This study examined the relationship between digital economy development and manufacturing carbon emissions in nine PRD cities from 2011 to 2024. Using fixed-effects and spatial econometric models, mediation analysis, decoupling measures, and regional subgroup regressions, the study investigated the overall association, potential mechanisms, spatial dependence, and differences between core and peripheral cities.

The baseline SDM produced a negative coefficient for digital economy development (-0.3423, $p < 0.05$). Negative and significant coefficients were also obtained when the spatial weight matrix was changed and when the 2024 observations were excluded; under the carbon-intensity specification the coefficient was positive and statistically insignificant.

In the non-spatial mediation model, the indirect association involving the secondary-industry share represented 29.82% of the estimated total association. The corresponding indirect association involving R&D expenditure was not statistically significant. The spatial indirect effect was also statistically insignificant, providing no clear evidence of a measurable cross-city emission response associated with digital economy development.

The subgroup regressions showed different coefficient patterns across the two groups. The digital economy coefficient was negative and statistically significant in the four core cities but positive and insignificant in the five peripheral cities. These findings suggest that the relationship between digital development and manufacturing emissions varies with regional industrial conditions rather than following a uniform pattern.

The study contributes evidence on the roles of industrial composition and regional heterogeneity in the relationship between digital economy development and manufacturing carbon performance. For manufacturing-intensive urban regions, the findings support differentiated digital development strategies, closer integration of industrial and carbon policies, and regional accounting arrangements that distinguish genuine emissions reductions from changes in the location of production.

Author Contributions

Conceptualization, Y.H.Y.; methodology, Y.H.Y.; validation, Y.H.Y.; formal analysis, Y.X.; investigation, Y.X. and D.L.T.; data curation, Y.X.; writing—original draft preparation, Y.X.; writing—review and editing, Y.H.Y. and D.L.T.; visualization, D.L.T.; supervision, Y.H.Y. All authors have read and agreed to the published version of the manuscript.

Funding

This study is supported by the Humanities and Social Sciences Foundation of the Ministry of Education of China (Grant No.: 24YJC630273); the Philosophy and Social Science Foundation of Guangdong Province (Grant No.: GD24CYJ48); the Foshan Philosophy and Social Science Planning Project (Grant No.: 2026-ZDB08); and the Jiangmen Philosophy and Social Science Planning Project (Grant No.: JM2026C17).

Data Availability

The data used in this study are compiled from publicly available statistical yearbooks and statistical communiqués. The compiled panel and the estimation code are available from the corresponding author on reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] W. Zhang, X. Liu, D. Wang, and J. Zhou, "Digital economy and carbon emission performance: Evidence at China's city level," *Energy Polic.*, vol. 165, p. 112927, 2022. https://doi.org/10.1016/j.enpol.2022.112927

[2] Q. Ma, M. Tariq, H. Mahmood, and Z. Khan, "The nexus between digital economy and carbon dioxide emissions in China: The moderating role of investments in research and development," *Technol. Soc.*, vol. 68, p. 101910, 2022. https://doi.org/10.1016/j.techsoc.2022.101910

[3] B. Xie, R. Liu, and R. Dwivedi, "Digital economy, structural deviation, and regional carbon emissions," *J. Clean. Prod.*, vol. 434, p. 139890, 2024. https://doi.org/10.1016/j.jclepro.2023.139890

[4] Z. Li, X. Chen, Y. Ye, F. Wang, K. Liao, and C. Wang, "The impact of digital economy on industrial carbon emission efficiency at the city level in China: Gravity movement trajectories and driving mechanisms," *Environ. Technol. Innov.*, vol. 33, p. 103511, 2024. https://doi.org/10.1016/j.eti.2023.103511

[5] Z. Li and J. Wang, "The dynamic impact of digital economy on carbon emission reduction: Evidence city-level empirical data in China," *J. Clean. Prod.*, vol. 351, p. 131570, 2022. https://doi.org/10.1016/j.jclepro.2022.131570

[6] W. Du, X. Liu, Y. Liu, and J. Xie, "Digital economy and carbon emission efficiency in three major urban agglomerations of China: A U-shaped journey towards green development," *J. Environ. Manag.*, vol. 373, p. 123571, 2025. https://doi.org/10.1016/j.jenvman.2024.123571

[7] Q. Xu, M. Zhong, and M. Cao, "Does digital investment affect carbon efficiency? Spatial effect and mechanism discussion," *Sci. Total Environ.*, vol. 827, p. 154321, 2022. https://doi.org/10.1016/j.scitotenv.2022.154321

[8] Y. Li, N. Li, and Z. Li, "Evolution of carbon emissions in China's digital economy: An empirical analysis from an entire industry chain perspective," *J. Clean. Prod.*, vol. 414, p. 137419, 2023. https://doi.org/10.1016/j.jclepro.2023.137419

[9] J. Sun and X. Wu, "Digital economy, energy structure and carbon emissions: Evidence from China," *Sage Open*, vol. 14, no. 2, p. 21582440241255756, 2024. https://doi.org/10.1177/21582440241255756

[10] D. Avom, H. Nkengfack, H. K. Fotio, and A. Totouom, "ICT and environmental quality in Sub-Saharan Africa: Effects and transmission channels," *Technol. Forecast. Soc. Change*, vol. 155, p. 120028, 2020. https://doi.org/10.1016/j.techfore.2020.120028

[11] P. Sadorsky, "Information communication technology and electricity consumption in emerging economies," *Energy Policy*, vol. 48, pp. 130–136, 2012. https://doi.org/10.1016/j.enpol.2012.04.064

[12] T. T. L. Chong, S. Wang, and C. Zhang, "Understanding the digital economy in China: Characteristics, challenges, and prospects," *Econ. Polit. Stud.*, vol. 11, no. 4, pp. 419–440, 2023. https://doi.org/10.1080/20954816.2023.2269327

[13] Z. Jin, H. Wang, C. Luo, and C. Y. Guo, "The impact of digital transformation on international carbon competitiveness: Empirical evidence from manufacturing decomposition," *J. Clean. Prod.*, vol. 443, p. 141184, 2024. https://doi.org/10.1016/j.jclepro.2024.141184

[14] S. He, S. Xu, and A. X. Zou, "Does green technological diversification impact industrial carbon emissions efficiency? The role of technological specialisation," *J. Innov. Knowl.*, vol. 10, no. 3, p. 100730, 2025. https://doi.org/10.1016/j.jik.2025.100730

[15] W. S. Chen and Y. T. Jiang, "Effect of digital economy development on carbon emission intensity: Evidence from Chinese provinces," *Front. Environ. Sci.*, vol. 13, p. 1618432, 2025. https://doi.org/10.3389/fenvs.2025.1618432

[16] B. R. Copeland and M. S. Taylor, "North-South trade and the environment," *Q. J. Econ.*, vol. 109, no. 3, pp. 755–787, 1994. https://doi.org/10.2307/2118421

[17] G. M. Grossman and A. B. Krueger, "Economic growth and the environment," *Q. J. Econ.*, vol. 110, no. 2, pp. 353–377, 1995. https://doi.org/10.2307/2118443

[18] D. Acemoglu, P. Aghion, L. Bursztyn, and D. Hemous, "The environment and directed technical change," *Am. Econ. Rev.*, vol. 102, no. 1, pp. 131–166, 2012. https://doi.org/10.1257/aer.102.1.131

[19] P. Tapio, "Towards a theory of decoupling: Degrees of decoupling in the EU and the case of road traffic in Finland between 1970 and 2001," *Transp. Policy*, vol. 12, no. 2, pp. 137–151, 2005. https://doi.org/10.1016/j.tranpol.2005.01.001

[20] C. Freitag, M. Berners-Lee, K. Widdicks, B. Knowles, G. S. Blair, and A. Friday, "The real climate and transformative impact of ICT: A critique of estimates, trends, and regulations," *Patterns*, vol. 2, no. 9, p. 100340, 2021. https://doi.org/10.1016/j.patter.2021.100340

[21] E. Masanet, A. Shehabi, N. Lei, S. Smith, and J. Koomey, "Recalibrating global data center energy-use estimates," *Science*, vol. 367, no. 6481, pp. 984–986, 2020. https://doi.org/10.1126/science.aba3758

[22] J. Li, Y. Wang, S. Zhang, and X. Sheng, "Digital economy, fiscal decentralization, and carbon emission intensity: Evidence from China," *Sustain. Futures*, vol. 9, p. 100522, 2025. https://doi.org/10.1016/j.sftr.2025.100522

[23] S. Ding, S. Wei, H. Zhang, and J. Hu, "Promotion or inhibition? Unveiling the heterogeneous effects and mechanism of the digital economy on carbon emission efficiency," *Energy Econ.*, vol. 149, p. 108803, 2025. https://doi.org/10.1016/j.eneco.2025.108803

[24] W. Xia, Z. Ruan, S. Ma, J. Zhao, and J. Yan, "Can the digital economy enhance carbon emission efficiency? Evidence from 269 cities in China," *Int. Rev. Econ. Finance*, vol. 97, p. 103815, 2025. https://doi.org/10.1016/j.iref.2024.103815

[25] Z. Ma, H. Xiao, J. Li, H. Chen, and W. Chen, "Study on how the digital economy affects urban carbon emissions," *Renew. Sustain. Energy Rev.*, vol. 207, p. 114910, 2025. https://doi.org/10.1016/j.rser.2024.114910

[26] Y. Ma and R. Li, "The impact of digital economy on carbon emissions: Insights from the G-20 energy transition and environmental governance," *Energy Strategy Rev.*, vol. 57, p. 101612, 2025. https://doi.org/10.1016/j.esr.2024.101612

[27] J. Hong, H. Huang, X. Wang, B. Dockerill, J. Ye, and S. Zhang, "Structural effects of provincial digital economy on carbon emissions within China: A multi-region input-output based structural decomposition analysis," *Sci. Total Environ.*, vol. 934, p. 173140, 2024. https://doi.org/10.1016/j.scitotenv.2024.173140

[28] Y. Chen and J. Liu, "Digital economy and carbon emissions: Spatial spillover effect and industrial structure mediation effect in China," *Technol. Econ. Dev. Econ.*, vol. 31, no. 3, pp. 793–818, 2025. https://doi.org/10.3846/tede.2025.22955

[29] H. Zhang, R. Qi, Y. Liu, T. Wang, F. Zhong, Q. Zhou, and K. Wang, "The spatial impact of digital economy on carbon emissions reduction: Evidence from 215 cities in China," *Front. Environ. Sci.*, vol. 12, p. 1370938, 2024. https://doi.org/10.3389/fenvs.2024.1370938

[30] J. Hou and Y. Fu, "Study on the spillover effect of digital economy development on CO_2 emissions," *Front. Energy Res.*, vol. 12, p. 1358093, 2024. https://doi.org/10.3389/fenrg.2024.1358093

[31] S. Ren, Y. Hao, L. Xu, H. Wu, and N. Ba, "Digitalization and energy: How does internet development affect China's energy consumption?" *Energy Econ.*, vol. 98, p. 105220, 2021. https://doi.org/10.1016/j.eneco.2021.105220

[32] S. Luo, N. Yimamu, Y. Li, H. Wu, M. Irfan, and Y. Hao, "Digitalization and sustainable development: How could digital economy development improve green innovation in China?" *Bus. Strategy Environ.*, vol. 32, no. 4, pp. 1847–1871, 2023. https://doi.org/10.1002/bse.3223

[33] Y. Lyu, W. Wang, Y. Wu, and J. Zhang, "How does digital economy affect green total factor productivity? Evidence from China," *Sci. Total Environ.*, vol. 857, p. 159428, 2023. https://doi.org/10.1016/j.scitotenv.2022.159428

[34] Y. Hao, Y. Li, Y. Guo, J. Chai, C. Yang, and H. Wu, "Digitalization and electricity consumption: Does internet development contribute to the reduction in electricity intensity in China?" *Energy Policy*, vol. 164, p. 112912, 2022. https://doi.org/10.1016/j.enpol.2022.112912

[35] X. J. Deng, M. Q. Yuan, and C. Y. Luo, "Corporate digital transformation, market competition, and the environmental performance—Microevidence from Chinese manufacturing," *Bus. Strategy Environ.*, vol. 33, no. 4, pp. 3279–3298, 2024. https://doi.org/10.1002/bse.3660

[36] Y. Shen, G. Wang, X. Wu, and C. Shen, "Digital economy, technological progress, and carbon emissions in Chinese provinces," *Sci. Rep.*, vol. 14, p. 23001, 2024. https://doi.org/10.1038/s41598-024-74573-4

[37] J. LeSage and R. K. Pace, *Introduction to Spatial Econometrics*. Chapman and Hall/CRC, 2009.

[38] J. Li, L. Ji, S. Zhang, and Y. Zhu, "Digital technology, green innovation, and the carbon performance of manufacturing enterprises," *Front. Environ. Sci.*, vol. 12, p. 1384332, 2024. https://doi.org/10.3389/fenvs.20

24.1384332
[39] K. Feng, S. J. Davis, L. Sun, X. Li, D. Guan, W. Liu, Z. Liu, and K. Hubacek, "Outsourcing CO_2 within China," *Proc. Natl. Acad. Sci.*, vol. 110, no. 28, pp. 11 654–11 659, 2013. https://doi.org/10.1073/pnas.1219918110
[40] S. J. Davis and K. Caldeira, "Consumption-based accounting of CO_2 emissions," *Proc. Natl. Acad. Sci.*, vol. 107, no. 12, pp. 5687–5692, 2010. https://doi.org/10.1073/pnas.0906974107
[41] X. Hao, Y. Li, S. Ren, H. Wu, and Y. Hao, "The role of digitalization on green economic growth: Does industrial structure optimization and green innovation matter?" *J. Environ. Manage.*, vol. 325, p. 116504, 2023. https://doi.org/10.1016/j.jenvman.2022.116504
[42] K. Imai, L. Keele, and D. Tingley, "A general approach to causal mediation analysis," *Psychol. Methods*, vol. 15, no. 4, pp. 309–334, 2010. https://doi.org/10.1037/a0020761
[43] Z. Wen and B. Ye, "Analyses of mediating effects: The development of methods and models," *Adv. Psychol. Sci.*, vol. 22, no. 5, pp. 731–745, 2014. https://doi.org/10.3724/sp.j.1042.2014.00731
[44] S. Lange, J. Pohl, and T. Santarius, "Digitalization and energy consumption. Does ICT reduce energy demand?" *Ecol. Econ.*, vol. 176, p. 106760, 2020. https://doi.org/10.1016/j.ecolecon.2020.106760
[45] Z. Ma, X. Niu, W. Ma, Y. Kang, and W. Xie, "Interregional and intersectoral interaction of digital economy in China," *Front. Environ. Sci.*, vol. 11, p. 1138200, 2023. https://doi.org/10.3389/fenvs.2023.1138200
[46] IPCC, "2006 IPCC guidelines for national greenhouse gas inventories: Volume 2: Energy," Institute for Global Environmental Strategies, 2006. https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html
[47] T. Zhao, Z. Zhang, and S. Liang, "Digital economy, entrepreneurship, and high-quality economic development: Empirical evidence from urban China," *Manag. World*, vol. 36, no. 10, pp. 65–76, 2020. https://doi.org/10.19744/j.cnki.11-1235/f.2020.0154

Appendix

Data sources and processing record.

The variables used in the analysis were constructed from the sources listed below. All series cover the nine mainland cities of the Greater Bay Area over 2011–2024 (126 observations).

Variable	Source	Construction and Processing
Manufacturing carbon emissions (E)	Authors' calculation from official statistics	CO_2 = manufacturing output value × comprehensive energy consumption per unit of output × 0.7476 t C/tce × 44/12; 10^4 tonnes of CO_2
Manufacturing output value (GM)	Municipal statistical yearbooks and communiqués	Gross manufacturing output 2011–2019; above-designated-size gross industrial output 2020–2024; 2024 estimated from reported growth; constant 2011 prices, 10^8 CNY
Energy consumption per unit of output (EC)	Provincial and municipal energy statistics	2005 constant-price basis; 2020–2024 extrapolated from provincial reduction targets and city documentation; tce per CNY 10,000
Digital economy index (DE)	Authors' calculation; Peking University Digital Financial Inclusion Index	Five indicators; annual min–max normalization; annual entropy weights; zeros replaced by 10^{-5}; 2020 from interpolated series; 2024 recalculated on the national sample
Per-capita GDP (PGDP)	Municipal statistical yearbooks	Constant 2011 prices; natural logarithm
Foreign direct investment (FDI)	Municipal statistical yearbooks	Utilized FDI, 10^4 USD; natural logarithm
Industrial structure (IND)	Municipal statistical yearbooks	Secondary-industry value added as a share of GDP, %
R&D expenditure (RD)	Municipal statistical yearbooks	10^4 CNY; natural logarithm
Energy intensity change (EI)	Provincial and municipal energy statistics	Annual percentage change in energy consumption per unit of output

Annual precipitation (RA)	Meteorological statistical records	Millimetres; natural logarithm

Note: Missing cell values were interpolated linearly within each city where adjacent observations existed; where interpolation was not feasible, values were estimated from the growth rates of comparable cities in Guangdong Province. The 2020 values of the digital economy index were taken from the linearly interpolated source series, and the 2024 values were recalculated on the national city sample.